Statistical Analysis

H_0 ✓
H_1 ✗

i) Z-test } Z-table → Z score
ii) t test } t table → t score
iii) chi-square ⇒ categorical data (Gender, 0/1)
iv) ANOVA ⇒ Variance

Z-test ⇒ we have Population std (σ) ii) n ≥ 30

Eg:- $\mu = 168$ cm ($\sigma = 3.9$)

i) The average height of all residents in a city is 168 cm. A doctor (L WOODA) believes the Mean to be different. He Measured the height of 36 individuals and found the average height to be 169.5 cm

a) State null and Alternative hypothesis
b) At a 95% Confidence level, is there Enough Evidence to reject the null hypothesis

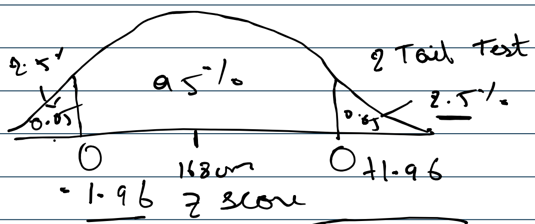
$\mu = 168$ cm → population mean
$\sigma = 3.9$ $n = 36$ $\bar{x} = 169.5$ cm → Sample mean

$CI = 0.95$ $\alpha = 1 - CI \Rightarrow 1 - 0.95 \Rightarrow 0.05$ $\boxed{\alpha = 0.05}$

i) Null hypothesis $H_0 = \mu = 168$ cm

ii) Alternative hypothesis $H_1 = \mu \neq 168$ cm

iii) Based on C.I



$0.025 \Rightarrow 1 - 0.025 = \boxed{0.9750}$ | +1.96

if Z is less than −1.96 or greater than 1.96, Reject the Null hypothesis.

$-1.96 < Z > 1.96$ Reject H_0 ✗ H_1 ✓

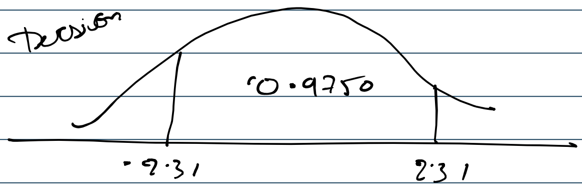
Z-test

$$Z_d = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{169.5 - 168}{3.9 / \sqrt{36}} = \frac{1.5}{0.065} = 2.31$$

$2.31 > 1.96$ Reject the Null hypothesis

$p < 0.05$

Decision



$1 - 0.9750$

$\Rightarrow 0.014$

$p = 0.01044 + 0.01044$

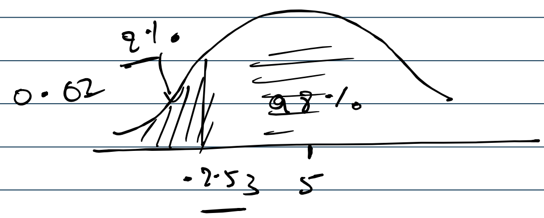
$= 0.02088$

$p < 0.05$

Rej

(2) A bed Manufacturer test all the bed after testing 40 beds the avg years they last with out damage is 4.8 year the Company 5 year (σ 0.50)

Ans)



$$Z\text{-test} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{4.8 - 5}{0.50 / \sqrt{40}} = -2.53$$

Area under the curve $\Rightarrow 0.00570$

$pvalue = 0.00570$

$0.00570 < 0.02$

we accept null hypothesis